



Spring Framework

IoC · AOP · MVC · Security · Events · Testing

Overview

Spring Boot 3 / Spring Framework 6. Every sub-chapter maps to named beans, configurations, and tests in ShopVerse.

IoC & Bean Lifecycle

Feature	Class / Annotation	Detail
Constructor injection	Every @Service, @Component, @Repository	Not @Autowired on fields
@Qualifier	primaryDataSource / readDataSource	Two DataSource beans
@Primary	Routing DataSource	Routes to primary or replica
@PostConstruct	KafkaProducerBean	Warms up producer on startup
@PreDestroy	KafkaProducerBean	Flushes and closes producer
Request scope	RequestScopedAuditContext	Holds traceId; injected via ObjectProvider
Prototype scope	OrderBuilder bean	New instance per injection point
InitializingBean	ApplicationReadinessValidator	Validates env vars on startup

AOP Aspects

Annotation	Advice Type	Behaviour
@LogExecutionTime	@Around	Logs method name, duration, arguments
@RateLimit(perMinute=60)	@Around	Redis INCR/EXPIRE per user
@Auditable	@AfterReturning	Writes audit entry for annotated service methods
@ValidateInput	@Before	Runs bean validation on method arguments

Note: CGLIB proxy for classes; JDK dynamic proxy for interfaces.
@EnableAspectJAutoProxy(proxyTargetClass=true)

Spring Security Filter Chain

SecurityFilterChain

CORS Filter

CSRF Filter (disabled for REST; enabled for admin UI)

JwtAuthenticationFilter extends OncePerRequestFilter

JwtTokenProvider (issue + validate HS512, 15 min access /

CustomerUserDetailsService

Authorization Rules

@PreAuthorize("hasRole('ADMIN')") → ProductController.delete

@PostAuthorize("returnObject.customerId == authentication.na

Transaction Propagation Summary

Method	Isolation / Propagation	Reason
OrderApplicationService.placeOrder()	REPEATABLE_READ + REQUIRED	Prevent concurrent oversell
PaymentService.charge()	REQUIRES_NEW	Payment failure rolls back only payment
AuditService.write()	NOT_SUPPORTED	Always runs outside caller transaction
ReportService.generateDailyReport()	SERIALIZABLE	Consistent snapshot of all orders

Testing Slice Summary

Annotation	Scope	Key Use
@ExtendWith(MockitoExtensionClass)	Unit	Every service class
@WebMvcTest(OrderController.class)	MockMvc slice	MockMvc + @MockBean for services
@DataJpaTest	JPA slice	In-memory H2; real repository queries
@SpringBootTest(RANDOM_PORT)	Full integration	TestRestTemplate
@Container (Testcontainers)	Integration	PostgreSQL, Redis, Kafka containers

Event Flow

OrderApplicationService publishes OrderPlacedEvent

@EventListener → InventoryEventListener (synchronous)

@Async @EventListener → NotificationEventListener (async,

@TransactionalEventListener(AFTER_COMMIT) → SearchIndexLis